Ethereum Network Activity & User Behaviour Analysis

Project Overview

This project analyzes how people are using the Ethereum network over time. The goal is to understand:

- Are more people using the network?
- How active are users?
- Are costs (fees) affecting user behaviour?

To answer these questions, I have built a dashboard using data from Dune and visualized it in Tableau.

Why This Project Matters

Ethereum is one of the most widely used blockchain networks. By analyzing its activity, we can understand:

- User growth and adoption
- How people interact with the network
- Whether usage is increasing or slowing down

This type of analysis is valuable for:

- Investors
- Product teams
- Blockchain companies

Tools & Data Used

- **Data Source:** Dune (blockchain data platform)
- **Visualization Tool:** Tableau
- **Data Pipeline:** Dune → Google Sheets → Tableau

What Was Analyzed

The dashboard focuses on four key areas:

1. **Network Activity**
   It helps understand the overall usage of the network. It includes two key indicators of block chain activity:
   - Total number of transactions
   - Total value transferred (ETH)

2. **User Growth**
   It shows whether more people are joining or leaving the network. It includes:
   - Number of active users (wallets)

3. **User Behaviour**
   It helps understand user engagement and participation patterns. It includes:
   - New vs returning users

4. **Network Cost**
   It helps understand the cost of using the network and impact of fee changes on user activity. It includes:
   - Average transaction fee (gas fee)

Key Insights

Daily Transactions — Analysis

Over the roughly 4-month period analysed (January to May 2026), Ethereum processed over **202 million transactions** in total, averaging around **2.26 million transactions every single day**. Day-to-day volumes stayed within a consistent range, which tells us the network has a **reliable, steady base of usage**.

The most notable pattern is a **burst of elevated activity throughout April**, where daily transactions climbed to around 3.5M on two separate occasions (around April 12 and April 28). This kind of double-peak within the same month suggests April was genuinely a busier period for the network, likely triggered by broader market activity or increased usage of Ethereum-based applications.

Toward the end of the period, volumes appear to ease slightly, though they remain well within the normal range seen throughout the analysis. Overall, the transaction data reflects a network that is **actively and consistently used**, with occasional surges that follow broader activity in the crypto market.

Daily Active Users — Analysis

Active users tell a more concerning story. The period peaked at **1.19M users on February 1**, but by early May had fallen to a low of **381K**, a drop of nearly 68% from peak.

Comparing the first and last 30 days of the period directly, average daily active users declined from **826,885 to 523,204, a fall of 36.7%**.

This divergence from the relatively stable transaction count suggests that while fewer wallets are active, those that remain are transacting more frequently, a sign of a consolidating but still engaged core user base.

New vs Returning Users — Analysis

Retention is the strongest signal in the dashboard. Returning users account for **65.7% of all user activity** across the period (52.3M vs 27.3M), and on an average day, returning users make up **65.8%** of the active user base, ranging from a low of 50.6% to a high of 78.3%.

This consistently dominant returning user share indicates a loyal, habitual user base rather than one driven by one-off activity. Notably, during the activity spikes visible in late February and mid-March, both new and returning users rose together, suggesting those events were broad enough to attract and re-engage users simultaneously.

Average Gas Fees — Analysis

Gas fees are the cost a user pays to complete a transaction on Ethereum, think of it like a processing fee, paid in ETH. For most of the period, these fees were **low and affordable**, averaging around **0.0000725 ETH** on a typical day, making the network relatively cheap to use.

The data tells a clear two-chapter story. For roughly the **first 3 months**, fees stayed consistently low and stable, hovering mostly between **0.0000292 ETH and 0.0001000 ETH**, meaning users were transacting without meaningful cost pressure. Then in **mid-to-late April**, fees surged dramatically. The single most expensive day peaked at **0.0007302 ETH**, over **10 times the typical daily average**, and this wasn't a one-day blip. A total of **12 days** crossed significantly above normal, all clustered in April, meaning users faced elevated costs for nearly two consecutive weeks.

This spike directly aligns with the **transaction peaks seen around April 12 and April 28**. When more people compete to get their transactions processed simultaneously, fees get bid up, exactly what happened here. By May, fees had retreated back toward **0.0000470–0.0000680 ETH**, confirming the congestion was **temporary and event-driven** rather than a permanent shift in network costs.

Total ETH Transferred — Analysis

ETH transfer volumes are the most volatile metric in the dashboard, ranging from a low of **432K ETH to a peak of 5.21M ETH on February 6** — a 12x range.

Unlike transaction count which stays relatively stable, transfer volumes fluctuate dramatically day to day, pointing to the influence of large institutional transfers or whale activity rather than retail-driven flows.

Despite this volatility, the average of **1.69M ETH/day** across the period confirms sustained capital movement through the network.

Overall Conclusion

The four-month analysis of Ethereum (January to May 2026) reveals a network that is **stable and well-retained, but facing a gradual decline in active participation**.

Transaction volumes averaged 2.26 million per day across 121 days, remaining consistent throughout, with a notable burst of elevated activity in April. This steadiness signals a reliable base of usage that persists even during quieter market periods.

Retention is the standout finding. Returning users accounted for **65.7% of all activity** across the period, meaning Ethereum's existing user base keeps coming back without needing to be replaced by new users. However, daily active users dropped by **36.7%** comparing the first and last 30 days, a trend that warrants close monitoring going forward.

Gas fees stayed affordable for most of the period at around **0.0000725 ETH** on a typical day, with a temporary spike in late April reaching **0.0007302 ETH** that coincided with the transaction peaks before quickly resolving.

Overall, Ethereum shows the hallmarks of a **mature and resilient network**, with strong user loyalty and stable throughput. The key challenge ahead is reversing the decline in active users, as sustained growth will require bringing new participants into the ecosystem, not just retaining existing ones.

What I Learned

Through this project, I developed skills in:

- Analyzing blockchain transaction data
- Building automated data pipelines
- Creating dashboards in Tableau
- Turning raw data into meaningful insights

Future Improvements

If expanded further, this analysis could include:

- Tracking large investors ("whales")
- Analyzing token flows (buy/sell behavior)
- Studying specific DeFi platforms like Uniswap

Final Note

This project demonstrates my ability to:

- Work with real blockchain data
- Build automated dashboards
- Communicate insights clearly to both technical and non-technical audiences